2.1 The Periodic Table: Chemical Periodicity

Question Paper

Course	CIE A Level Chemistry
Section	2. Inorganic Chemistry
Topic	2.1 The Periodic Table: Chemical Periodicity
Difficulty	Hard

Time allowed: 20

Score: /10

Percentage: /100

Question 1

A student investigated the chloride of a Period 3 element. Below is a cutting from their lab book describing experiments and observations.

> The compound was a white crystalline solid. It dissolved easily in water to give a solution of pH 12. When placed in a test-tube and heated in a roaring Bunsen flame, the compound melted after several minutes of heating.

What can be deduced from this record?

A. At least one of the recorded observations is incorrect.

B. The compound was sodium chloride, $NaCl$.

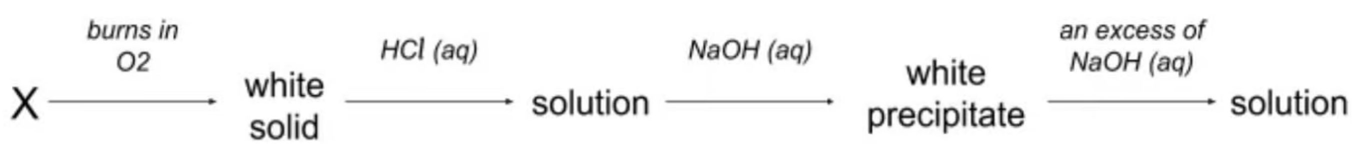
C. The compound was silicon tetrachloride, $SiCl_4$.

D. The compound was phosphorus pentachloride, PCl_5.

[1 mark]

Question 2

An element **X** reacts according to the following sequence.



What could be element **X**?

A. Mg

B. Si

C. Al

D. P

[1 mark]

Question 3

Use of the periodic table is relevant to this question.

Sir Humphrey Davy discovered the elements magnesium, boron, sodium and calcium.

Which of the elements Sir Davy discovered has the **third** lowest first ionisation energy in its period and the **third** smallest atomic radius in its group?

A. Magnesium

B. Boron

C. Sodium

D. Calcium

[1 mark]

Question 4

When solution **X** is titrated against hydrochloric acid, a neutralisation reaction occurs.

Which of the following hydroxides is not solution **X**?

A. Magnesium

B. Aluminium

C. Sodium

D. Phosphorous

[1 mark]

Question 5

G and J are oxides of different Period 3 elements.

If one mole of J is added to water, the solution formed is neutralised by exactly one mole of G.

What could be the identities of G and J?

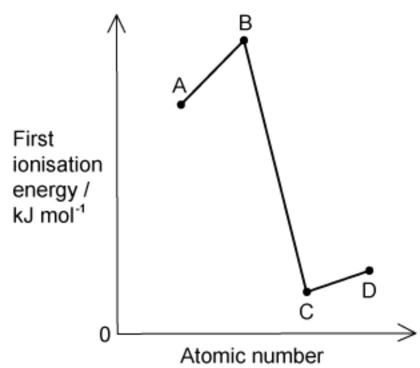
	G	J
A	Na_2O	SO_3
B	Na_2O	P_4O_{10}
C	Al_2O_3	SO_3
D	Al_2O_3	P_4O_{10}

[1 mark]

Question 6

Shown on the graph are the relative values of the first ionisation energies of four elements that have consecutive atomic numbers. One of the elements reacts with hydrogen to form a covalent compound with formula HX.

Which element could be X?



[1 mark]

Question 7

Mohr's salt is a pale green crystalline solid which is soluble in water. It contains two cations, of which one is Fe^{2+} and one anion which is SO_4^{2-}.

The second cation was determined by heating solid Mohr's salt with solid sodium hydroxide, and a colourless gas was evolved. The gas readily dissolved in water giving an alkaline solution.

A grey-green solid residue was also formed, which was insoluble in water.

Which row correctly identifies the colourless gas and the grey-green residue?

	Residue	**Gas**
A	$Fe(OH)_2$	NH_3
B	$Fe(OH)_2$	SO_2
C	Na_2SO_4	NH_3
D	Na_2SO_4	SO_2

[1 mark]

Question 8

Bacteria in swimming pools are killed by adding aqueous sodium chlorate (I), NaOC*l*. This reacts with water to produce HOC*l* molecules which can breach the bacteria cell wall and kill the bacteria. The reaction below shows the reaction with water:

$$OCl^-(aq) + H_2O \rightleftharpoons OH^-(aq) + HOCl(aq)$$

Under the sun's UV rays, the OCl^- ion is broken down by ultra-violet light:

$$OCl^-(aq) + \text{uv light} \rightarrow Cl^-(aq) + \tfrac{1}{2}O_2(g)$$

Which method would maintain the highest concentration of HOC*l* (aq)?

A. Add a solution containing chloride ions.

B. Acidify the pool water.

C. Bubble air through the water.

D. Add a solution of hydroxide ion.

[1 mark]

Question 9

The table gives the successive ionisation energies for an element *R*.

	1st	2nd	3rd	4th	5th	6th
Ionisation energy / kJ mol^{-1}	950	1800	2700	4800	6000	12300

What could be the oxidation state of element *R* in its chloride?

A. +1

B. +2

C. +3

D. +4

[1 mark]

Question 10

Which row describes a ceramic material?

	Conductivity of solid	Boiling point / K	Melting point / K
A	None	352	156
B	Good	1380	922
C	Good	2943	2130
D	None	3873	3125

[1 mark]

Model Answers: Hard

1

The correct answer is **A** because:

- The compound being investigated is **aluminium trichloride** ($AlCl_3$).
- $AlCl_3$ is an **ionic** compound with a **giant lattice** structure and is a **white crystalline solid** at room temperature.
- $AlCl_3$ **does not** dissolve to give a solution of pH 12, an ordinary solution of aqueous aluminium trichloride has a **pH of 2 – 3.**
  - $AlCl_3 (s) + 6H_2O (l) \rightarrow [Al(H_2O)_6]^{3+} (aq) + 3Cl^- (aq)$
  - $[Al(H_2O)_6]^{3+} (aq) \rightarrow [Al(H_2O)_5OH]^{2+} (aq) + H^+ (aq)$
- $AlCl_3$ melts at around 180 °C, which is lower than expected for an ionic lattice and is due to the bonds having a **slight covalent character.**

B is incorrect as **sodium chloride** gives a **neutral** solution when dissolved in water and has a **very high melting point** (801 °C) due to its **ionic lattice** structure.

C & **D** are incorrect as both $SiCl_4$ and PCl_5 are **liquids** at room temperature because they are **simple covalent** molecules and both produce **strongly acidic** solutions when added to water.

2

The correct answer is **C** because:

- When aluminium is burnt in oxygen, white **aluminium oxide** is formed:
  - $Al (s) + 3O_2 (g) \rightarrow 2Al_2O_3 (s)$
- Aluminium oxide can act as an acid and a base (is **amphoteric**), when added to hydrochloric acid a solution of **aluminium trichloride** is produced:
  - $Al_2O_3 (s) + 6HCl (aq) \rightarrow 2AlCl_3 (aq) + 3H_2O (l)$
- Adding sodium hydroxide to aluminium trichloride solution produces **insoluble aluminium hydroxide** and **soluble sodium chloride** solution:

- $AlCl_3$ (aq) + 3NaOH (aq) → $Al(OH)_3$ (s) + 3NaCl (aq)

- In an **excess** of sodium hydroxide the reaction continues and produces a **soluble, amphoteric compound**:
  - $Al(OH)_3$ (s) + NaOH (aq) → **$Na^+[Al(OH)_4]^-$** (aq)

A is incorrect as solid **$Mg(OH)_2$** made in the reaction of magnesium chloride and sodium hydroxide does **not** dissolve in an excess sodium hydroxide.

B & **D** are incorrect as silicon dioxide and phosphorus oxide are **not** able to react with hydrochloric acid because they have **no basic properties** and do **not** contain **oxide ions** (because they are **covalent** molecules).

3

The correct answer is **D** because:

- The first ionisation energy of an element is the energy required to remove one mole of electrons from one mole of gaseous atoms to produce one mole of gaseous ions: X (g) → X^- (g) + e^-.

- **Group 2 elements** have the 3rd lowest ionisation energy in their period because despite the overall increase in energy across a period group 3 elements have **lower** first ionisation energy than group 2 elements because the electron in the **p orbital** is more **shielded** and therefore the **nuclear pull** is reduced.

- Therefore, the element is either **magnesium or calcium.**

- The atomic radius is the distance between the nucleus and the outer electron orbital and increases down a group because an extra orbital is being added each time.

- Therefore the 3rd smallest atomic radius in a group would belong to the element in period 4, which in this case is **calcium.**

4

The correct answer is **D** because:

- To undergo a reaction with hydrochloric acid, the hydroxide must be able to act as a **base.**

- Phosphorous “hydroxide” is also known as **phosphoric acid** and has the formula H_3PO_4; it has a central phosphorus atom bonded to **3 hydroxide groups** and an **oxygen.**
- In solution phosphoric acid **weakly dissociates** into $H_2PO_4^-$ and H^+ however due to it only have one P=O double bond the charge is not easily stabilised.
- Therefore, phosphorous “hydroxide” **cannot** react with hydrochloric acid.

A is incorrect as a neutralisation reaction takes place: $Mg(OH)_2 + 2HCl \rightarrow MgCl_2 + 2H_2O$

B is incorrect as a neutralisation reaction takes place: $Al(OH)_3 + 3HCl \rightarrow AlCl_3 + 3H_2O$

C is incorrect as a neutralisation reaction takes place: $NaOH + HCl \rightarrow NaCl + H_2O$

5

The correct answer is **A** because:

- When $\mathbf{SO_3}$ is added to water, it produces **sulfuric acid** solution (pH 0 when concentrated):
  - $SO_3\,(l) + H_2O\,(l) \rightarrow H_2SO_4\,(aq)$
- When one mole of sulfuric acid is added to one mole of the **strongly basic** $\mathbf{Na_2O}$, the following **neutralisation** reaction takes place:
  - $H_2SO_4 + Na_2O \rightarrow Na_2SO_4 + H_2O$
- Sodium sulfate is a salt that forms neutral solutions of pH 7.

B is incorrect as one mole of phosphorous oxide makes **four** moles of phosphoric acid ($4H_3PO_4$) which is not neutralised by one mole of Na_2O.

C is incorrect as the final product would be **aluminum sulfate** which is slightly **acidic.**

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D is incorrect as one mole of phosphorous oxide makes **four** moles of phosphoric acid ($4H_3PO_4$) which is not neutralised by one mole of Al_2O_3.

6

The correct answer is **A** because:

- The 4 consecutive elements are from Group 7 (A), Group 8 (B), Group 1 (C), Group 2 (D).
- For an element to form a compound with hydrogen with the formula HX, we know it needs to have **7 electrons** in its outer ring (so it can bond with hydrogen to get the extra 1 electron needed for a full orbital) and therefore we know A is a **Group 7** element.
- There is a **general rise** in first ionisation energy as you go **across a period** because the **nuclear charge increases** and therefore the pull on the electrons increases.
- When electrons are added to a **new orbital** (at the start of a new period) the first ionisation energy **drops** drastically.
- This is because of:
  - the increase in **shielding,** (due to an extra orbital between the electron and the nucleus).
  - the increased **distance** and therefore the **reduced attraction** to the nucleus

outweighs the pull from the extra proton.

7

The correct answer is **A** because:

- If NaOH reacted with the sulfate ion (SO_4^{2-}), then **sodium sulfate** (Na_2SO_4) would be made, which is a **solid that dissolves** into water producing a **neutral** solution.
- Therefore, the gas can be assumed to be **NH_3** shown by the half-equation:
  - $NH_4^- + OH^- \rightarrow NH_3 + H_2O$
- The remaining ions are Na+ from the sodium hydroxide and Fe^{2+} and SO_4^{2-} from Mohr's salt as well as hydroxide ions.
- **Sodium sulfate** (Na_2SO_4) is a **white crystalline solid** with **ionic** bonding and is therefore **soluble** in water.
- **Iron hydroxide** ($Fe(OH)_2$) is a **transition metal hydroxide** and therefore **insoluble**, this is due to their very high ionisation energies.

8

The correct answer is **B** because:

- Increasing the pool water acidity will **reduce** the **concentration of OH^- ions** and therefore cause the **equilibrium to shift** in favour of the **forward** reaction.
- The equilibrium shift will opposite the change in OH^- concentration and **produce more OH^- ions,** and therefore **more HOC*l*.**

A is incorrect as adding more chloride ions **won't** affect the equilibrium because there are no chloride ions in the reaction.

C is incorrect as adding more air (O_2, CO_2 and N_2) **won't** affect the equilibrium because none of these compounds is in the reaction.

D is incorrect as adding hydroxide ions will shift the **equilibrium to the left**, favouring the **backward** reaction in order to **oppose the increase** in OH^- concentration. This would, therefore **reduce** the concentration of HOC*l*.

9

The correct answer is **C** because:

- The biggest difference is between the **5th and 6th** ionisation energies.
- This means removing the 6th electron will be the most difficult, implying element R has **5 electrons in its outer shell.**
- Removing an electron from the orbital lower that the out orbital will be much harder due the reduced **distance** from the nucleus and less **shielding**, meaning the electron is **more attracted** to the nucleus.
- Elements with 5 electrons in their outer shell are in **group 5**, and will therefore bond with **3 chlorine atoms** – this will result in all atoms having 8 electrons in their outer shell.
- Each chlorine atom has an oxidation number of -1. Therefore element R must have an oxidation number equal and opposite to 3 chlorine atoms (= 3 x 1) = **+3**

10

The correct answer is **D** because:

- Ceramic materials are **crystalline oxides**, **nitrides** and **carbides.**
- Ceramic materials are **solid,** and regardless of whether the bonds are ionic or covalent, **giant lattice** structures are formed.
- Giant lattice structures have **high** melting and boiling points because strong ionic or covalent bonds must be broken in order to separate individual compounds.
- Giant lattice structures also **do not** conduct electricity when in the solid state